

Navigating Afrocentric Human-Computer Interaction Research: A Scoping Review and Proposition of Afro-Postmodernism for Decolonial Praxis

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ABSTRACT

Ongoing Human-Computer Interaction (HCI) studies in Africa have revealed the prevalence of colonial influences in current research and design practices. These colonial influences have given rise to racism and discriminatory practices in research, consequently affecting study outcomes and design implications. Responding to this, Afrocentricity which values critical thinking, cultural representation and inclusivity of African heritage has been explored. Its exclusive use, however, could increase the risk of essentialism and exclude Africans without indigenous cultural experiences. Using a keyword-based approach targeting HCI research publications, this scoping review examines 16 empirical HCI publications from 2013 to 2023. By assessing the impact of Afrocentricity on research and design practices, we identify ambiguities in the positionalities of researchers and its role in study design. To address these issues, we propose a deconstructive Afro-postmodernism framework.

CCS CONCEPTS

• **Human-centred computing** → **HCI theory, concepts and models.**

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1 INTRODUCTION

The impact of colonialism on different aspects of African societies, spanning socio-cultural and demographic dimensions, is an incontrovertible reality [86]. This historical phenomenon ushered in different paradigms of education [40, 47, 121], religious norms [59, 110], and political ideologies [6, 34], which, while possibly yielding certain advantages, concurrently brought about the marginalization of indigenous African cultures and propagated injurious ideologies such as racism [12, 58]. The first academic institutions of higher learning within colonized African communities were

grounded in Western pedagogical frameworks [79, 96], thereby resulting in a monolithic approach based on the Eurocentric perception of education. Till this day, most academic institutions have adopted Western models of teaching and learning. This trend has adversely impacted education, research and innovation in African communities [95], as they either deviate from or do not represent indigenous identities and lived experiences of Africans [90]. Having this in mind, Asante [16] championed Afrocentricity as a critical response to pervasive Westernised practices within academic frameworks. He sought to amend asymmetrical power dynamics that positioned African communities as objects of research. His advocacy was based on the premise that Africans had lost their sense of agency, were delusive and inherently unconfident as a result of colonialism. Numerous Human-Computer Interaction (HCI) researchers have embraced Asante's lens to understand socio-cultural nuances specific to the African experience and how they could influence research and design practices [20, 78, 81]. Nonetheless, this approach can be criticised as it has raised concerns related to epistemic relativism, the promotion of exclusionary dynamics and potential essentialism. The positionality of HCI researchers using this standpoint is therefore of paramount importance, as it could influence their approach to research and design practices, and therefore subsequent outcomes [15].

In this paper, we review 16 research papers with the primary aim of gaining comprehensive insights into the multifaceted perspectives of researchers concerning research and design practices when viewed through an Afro-centred lens. We evaluate current body of knowledge, arguing against the complete dismissal of certain colonial concepts. Despite their historical roots, these concepts must have been questioned, proven practical and necessary, and should be adapted to align with an inclusive, non-racist, and relevant African perspective. Importantly, we caution against the perils of oversimplification and distortion in the interpretation of the terms "Afrocentricity" or "Afrocentrism", recognizing the need to guard against gross misinterpretations that may arise from a lack of comprehensive understanding. In light of the developing presence of Afro-centred research in HCI, this study seeks to discern inherent limitations and provide recommendations that can serve as guiding principles for subsequent researchers.

2 RELATED WORK

2.1 The evolution of Colonialism in Africa

Decolonial literature argues that since the dissolution of European colonies in Africa, the educational system has significantly perpetuated colonial paradigms. Unfortunately, these paradigms are

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marked by cultural insensitivity, racial bias, and a lack of inclusivity [88, 100, 102]. Historically speaking, the stimulus for colonialism was not limited to the exploitation of natural resources, but the domination of culture [74, 77, 97] and psychological/political subjugation [100, 104, 105]. Superficially, socio-political incentives were reasons for colonialism, however, other factors relating to culture and behaviour began to manifest as the consequences of colonialism began to evolve [26]. In Africa, classical colonialism was symbolised by the apprehension and enslavement of Africans in Africa and the Caribbean [22, 56]. The transatlantic slave trade involved the transportation of people from Central and Western Africa to the Americas by European slave traders between the 16th and 19th century [92]. This resulted in the depopulation of the African continent [52, 57, 120] and socio-economic exploitation/total subjugation of affected African communities.

Religion played a huge role in the transatlantic slave trade, as some of the ruthless expeditions in Africa were founded on Christianity [28]. Indoctrination is said to have occurred simultaneously with slave voyages [112] and it is theorised that the superstitious nature of pre-colonial Africans created an avenue for religious infiltration by European colonizers [68]. Colonial religious influence altered pre-colonial African practices and ethics, leading to the elimination of perceived "barbaric" behaviors [94, 107, 127]. Although Christian missionaries had previously attempted to establish the religion in Africa several centuries before, the "Westernization" of Africa was significantly accomplished by their permeation in the 18th century [130]. This led to the conversion and transformation of African societies to Christian colonies with Western teaching practices [46, 70, 105]. Contrastingly, Islam was also introduced to several African societies by means of coercive colonial tactics [39, 80, 103]. While it is important to acknowledge that there were Africans who voluntarily converted to these religions (or others), the integration of non-African religions into the African society has often led to the development of syncretic belief systems that have, in some cases, overshadowed or suppressed authentic African cultures.

2.2 Decolonization, Afrocentricity, and their Intersection in HCI Research

Decolonization refers to the process of dismantling colonial systems and ideologies using a critical historical lens [19, 116]. It requires socio-economic, political, cultural and academic efforts to unravel and deconstruct colonial influences. Deconstruction is of paramount importance in decolonialist movements and in this context, it enables interested parties acknowledge the existence of nuances [62, 84, 115]. Decolonial researchers including those in the field of HCI say that the crux of decolonization is the elimination of direct/indirect control of institutions/practices by colonisers and concepts that perceive people as subordinate [20, 29, 63, 64, 81]. These approaches encourage researchers to question their modes of inquiry and research processes especially if they are founded on Eurocentric/colonial research criteria [113]. These criteria which may be biased, uninclusive, racist or dismissive, are inappropriate in a field like HCI where the study of human behaviour is significantly important [5]. Decolonisation has led to a resurgence of interest in African history, it has inspired expression and encouraged a

critical examination of colonial influences on African populations, therefore encouraging cultural and intellectual movements such as *Afrocentricity*. Afrocentricity by Asante [16] refers to a paradigm or concept that emphasizes the significance of ensuring African and African diasporic perspectives are centred culturally, academically, historically, economically and socially. It encourages the reorientation of Africans away from colonial (Eurocentric) influences. Asante, a prominent advocate of Afro-centred principles, expressed profound concern regarding the seemingly interchangeable use of the terms "*Afrocentricity*" and "*Afrocentrism*." In his seminal manifesto, he articulates a compelling argument where he posits that Afrocentricity has undergone a process of demonization, metamorphosing (mutating) into what he refers to as Afrocentrism [14].

His core contention resides in the nuanced distinction between these two terms, contending that Afrocentricity transcends mere "re-centring on blackness," assuming a more profound role of ensuring the authentic placement of Africans within their historical context and own time. From this foundational premise, the discourse pivots to the emergence of the argument that Africans have undergone a state of decentring and disorientation, a consequence of the conflated usage of these terms. Central to Asante's viewpoint is the assertion that Afrocentricity functions as an extension of Afrocentrism, encapsulating a broader, more encompassing framework. Within this expanded framework, not only are Africans, including those within the diaspora, enveloped but they are also nurtured to inhabit a space of centrality, both historically and culturally. However, in HCI and numerous other literature, the terms "Afrocentrism" and "Afrocentricity" have been used interchangeably [33, 131, 134]. Notably, certain researchers consistently opt for the use of the adjective "Afrocentric", without extensively probing into the nuances and potential ramifications stemming from the interchangeable use of these aforementioned terms. While "Afrocentric" as an adjective is "meant" to communicate the tenets of Afrocentricity and describe Afro-centred (focus on African/African diasporan identities) research, its vague use does not clarify the situatedness of a researcher. This trend questions the precise nature of their positionalities and intellectual orientations. Regardless, most HCI researchers have agreed that in order to tackle the issue of unbalanced cross-cultural representation in design practices, socio-cultural sensitivities and contexts must be recognised [43, 69].

3 METHOD

This scoping literature review synthesizes existing information and reveals different applications of Afrocentricity in HCI. Guidelines prescribed in the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) checklist were utilised in the conduction of this literature review [128]. Our selection process involved conducting a search for research articles on "Afrocentricity" in Human-Computer Interaction through extensively utilized academic databases that are known for retrieving HCI research. We do not conduct our search using keywords such as "decolonial" or "postcolonial", because although decolonization is an important complex process that inspires Afrocentricity, it inspires other perspectives such as subaltern studies, critical race theory, gender and queer studies, and anti-capitalist/anti-imperialist movements. Thus, its inclusion or use as the keyword for

this study will result in academic contributions, howbeit important, potentially out of the scope of this study.

3.1 Study Objectives

Given our curiosity about the application of Afrocentricity as a paradigm in HCI research, we organized our study around a carefully formulated research question. This question was designed to unveil epistemological complexities in Afro-centred approaches: *Through a critical lens, what is the positionality of researchers in relation to the diverse perspectives and narratives that encompass African lived experiences?*

3.2 Positionality

This review paper originates from the authors' deep passion for Afro-centred research, how it is influenced by researchers' positionalities and its relevance in HCI. Combining these areas of interest, we embark on an intellectual journey to discover how Afro-centred perspectives can enhance the understanding and design of interactive technologies. Based on our understanding of what Afro-centred research should be, we aim to synthesize existing knowledge, analyze relevant literature, and offer new insights, opening up exciting possibilities for Afro-centred approaches in HCI.

3.3 About the Analysts

The first author of this paper undertook the tasks of collecting relevant literature, conducting essential analysis, generating codes, presenting a balanced assessment of selected excerpts from the selected literature and proposing the Afro-postmodernist framework. Likewise, the second author analysed the extracted literature and inductively generated codes which were relevant to the objectives of the study. These codes were compared to those generated by the primary author, and final themes were generated. Finally, formal reviews and various contributions to different aspects of the paper were undertaken by the third author and external reviewers.

3.4 Strategy for Article Search

We explored 4 academic databases (ACM Digital Library, IEEE Xplore, SCOPUS (Elsevier) and Science Direct) to obtain relevant literature for the study. Article search was limited to empirical studies carried out in the past decade (2013-2023) due to an elevated interest in HCI decolonialist movements and the conduction of related research in low and middle income countries in Africa [79]. To identify relevant articles by searching through abstracts, titles and the body of work of these papers, we used "Afrocentric", "Afrocentrism" and "Afrocentricity" as the keywords. See **figure 1**.

3.4.1 Selection Criteria: The selection criteria for the critically analysed papers include:

- (1) They must have been written and published in English (or translated into English by the authors)
- (2) The reviewed literature must describe and/or imply the use of Afro-centred approaches in their research.
- (3) Only literature published within the past decade must be considered. Although this limits the number of contributions, studies with the most current evidence must be considered to make the review process easier.
- (4) Empirical literature must have been peer-reviewed and published in journals. Hence dissertations, incomplete literature, extended abstracts, pre-prints, review papers, and other grey literature are disqualified.

4 ANALYSIS AND RESULTS

4.1 Identification and Bibliographic data

- (1) In the ACM Digital Library, the selected keywords were inputted in the general search box, before narrowing to the "Journal" and "Conferences" sections. There were 20 results, out of which 2 articles were generated from the journal's section, while 18 were from published conference contributions. We were left with 11 research articles after we excluded extended abstracts from the ACM database.
- (2) The IEEE Xplore database generated 1 relevant research article
- (3) The search in the SCOPUS database was filtered for Computer Science and Engineering articles, and refined to include peer-reviewed literature. This generated 81 articles. After reviewing abstracts and other sections of the papers, 19 articles were removed because they did not align with the objectives of the study. Of the remaining 62 articles, only 2 were relevant to HCI.
- (4) Using appropriate filters in the Science Direct database generated 107 results. However, streamlining this search to Engineering and Computer Science subject areas resulted in 3 research articles. Out of which only two (2) were relevant in HCI.

4.2 Data Extraction and Analysis

Using Microsoft Excel, we systematically recorded significant information from each research paper, including details such as author names, publication title, publication year, and comprehensive abstracts. This approach ensured the methodical extraction and organization of relevant data for subsequent analysis. Thereafter, the research papers were independently analysed by the first and second author using abductive thematic analysis as inspired by Thompson [126] on NVivo 11. This approach was used to permit the researcher to establish specific parameters and limitations aligned with the study's goals [126]. It involves a satisfactory back and forth between inductive and deductive reasoning and requires the use of a code book. Ideas for analysis were influenced by several HCI contributions [37, 91, 114] and our approach was to begin with a reflexive inductive coding approach, followed by deductive iterations inspired by the researcher's positionality and peer-reviewed literature.

The primary author conducted two coding iterations starting with an inductive method of thematic analysis based on Braun and Clarke's reflexive approach [32]. The first and second coding iterations were based on the research objectives, however, the second iteration involved removing and/or merging generated codes. These iterations were conducted on the abstracts, introduction, discussion and conclusion of the reviewed papers. Some of the codes generated include *skin color*, *ethics* and *Eastern Africa*. On the other hand, using a similar approach, the second author generated codes

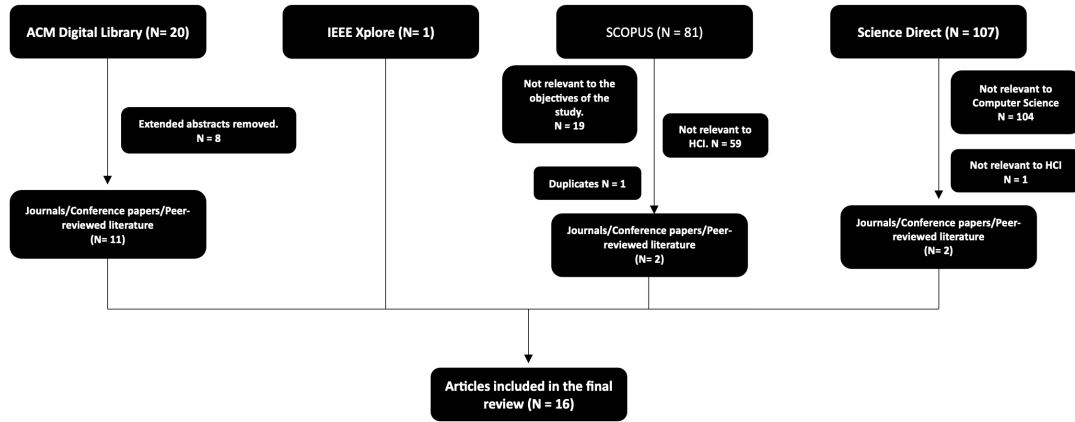


Figure 1: PRISMA flowchart describing the process of article selection

inductively and generated inferences based on existing scholarly evidence. Both researchers independently prepared a code book with three columns, the first being the generated code, the second being the coded section of the paper and the third being the rationale behind the coding selection. Thereafter, a virtual meeting between the primary and second author was held to comparatively analyse and reflect on the codes in both code books. Finally, a consensus was reached yielding three overarching themes that comprehensively encapsulate the research questions. Key information from the abstracted data are summarised in **Table 1**.

4.3 Generated Themes

We generated three themes that address Afro-centered challenges and communicate the situatedness of researchers within the reviewed HCI publications. It is important to note that although all papers align with a theme, they are not necessarily thematically exclusive. See **Figure 2**.

4.3.1 Artefacts inspired by Afrocentricity: Among the five research papers that offer insights into the influence of Afrocentric approaches on artefact design, three (**G**, **I**, **K**) aimed to address real-life challenges. Only one of them (**I**) explicitly identified Afrocentricity as its motivating paradigm. In general, the reviewed papers, except for **I**, were inspired by Afrocentricity but did not explicitly state their positionalities. The Afrocentric discourse ranged from solving racism/bias in design practices to disclosing design practices leveraging Afrocentric inspired methodologies. **A** [73] was based on the premise that an emotional response could be evoked on users as a result of colour. Using 175 participants, **A** explored this premise using robots. Findings from the study revealed that compared to white colored robots, darker (black and brown) robots were more likely to be attached to manual labour, and presumably stronger but fundamentally less intelligent. The

subterranean principle was that “skin color” which is directly associated with ethnicity is a perceptual bias that could significantly impact the design of artefacts. On the issue of race related bias, **G** [13] explored natural gestures of marginalized members of the San community in Namibia, studied their implementation in a Virtual Reality (VR) application and co-designed new gestures inspired by natural interactions. Both research papers provide clarity on the applicability of Afrocentric paradigms in robotics, human-factors and virtual reality.

“The results suggested that black robots were viewed as significantly stronger than yellow robots, that respondents were more likely to move away from black colorized robots, and that yellow robots were viewed as more affable than black robots” - **A** [73]

Upon reviewing the papers, we observed that **I** [27] and **K** [124] emphasized the significance of understanding the lived experience of Africans and involving them actively in design practices to achieve contextually suitable outcomes. Their respective research efforts was aimed at fostering adherence to local ethics and embracing indigenous practices, thereby facilitating a more culturally attuned design experience. While **I** delved into the relationship between culture, technology adoption, and future technological expectations, **K** centred their investigation on participatory design gaps concerning decolonialism, employing young Namibians as active collaborators. In parallel, **L** [72] presented a distinct endeavor wherein they employed an Afrocentric paradigm (once again, unclear) while crafting a specialized Pothole Detection, Classification, and Logging (PDCL) system tailored to accommodate the peculiarities of African roads. Collectively, these works ardently advocated for advancing HCI research through the promotion of inclusive and decolonised design practices. These approaches sought to be guided by a profound understanding of the cultural experiences of the African populace, ensuring that the technological solutions generated were truly resonant with their culture.

Table 1: An overview of the reviewed research articles using PRISMA-ScR guidelines

Article Identifier	Author names	Publication Title	Publication Year	Theme
A	Barfield [73]	Discrimination and Stereotypical Responses to Robots as a Function of Robot Colorization	2021	Artefacts inspired by Afrocentricity
B	Green and Chen [60]	Disparate Interactions: An Algorithm-in-the-Loop Analysis of Fairness in Risk Assessments	2019	Solving Real-Life Challenges Through Afrocentric-inspired Research
C	Awori et. al [17]	“It’s only when somebody says a tool worked for them that I believe it will work for me”: Socio-structure as a lens for digital transformation’	2022	Solving Real-Life Challenges Through Afrocentric-inspired Research
D	Amugongo et. al [9]	Invigorating Ubuntu Ethics in AI for health-care: Enabling equitable care	2023	Exploring Researcher Positionality and Cultural Interventions
E	Metfula and Chigona [87]	In search of development in a national ICT policy: The case of Swaziland	2014	Solving Real-Life Challenges Through Afrocentric-inspired Research
F	Mukumbira et. al [93]	Conservation tool design caught in the midst of transdisciplinary negotiations of data accuracy	2021	Solving Real-Life Challenges Through Afrocentric-inspired Research
G	Arendtthorp et. al [13]	Overcoming Legacy Bias: Re-Designing Gesture Interactions in Virtual Reality With a San Community in Namibia	2022	Artefacts inspired by Afrocentricity
H	Eglash et. al [45]	Counter-hegemonic Computing: Toward Computer Science Education for Value Generation and Emancipation	2021	Exploring Researcher Positionality and Cultural Interventions
I	Cabrero et. al [27]	A UX and Usability expression of Pastoral OvaHimba: Personas in the Making and Doing	2016	Artefacts inspired by Afrocentricity
J	Stichel et. al [124]	Namibian Indigenous Communities Reflecting on Their Own Digital Representations	2019	Exploring Researcher Positionality and Cultural Interventions
K	Smith [124]	Decolonizing Participatory Design: Memory Making in Namibia	2020	Artefacts inspired by Afrocentricity
L	Jordan et. al [72]	Road to Repair (R2R): An Afrocentric Sensor-Based Solution to Enhanced Road Maintenance	2022	Artefacts inspired by Afrocentricity
M	Jakubeit et. al [71]	Designing a problem analysis toolkit in the context of humanitarian engineering	2023	Solving Real-Life Challenges Through Afrocentric-inspired Research
N	Eyssel [51]	An experimental psychological perspective on social robotics	2016	Exploring Researcher Positionality and Cultural Interventions
O	Krauss [76]	Collisions between the Worldviews of International ICT Policy-Makers and a Deep Rural Community in South Africa: Assumptions, Interpretation, Implementation, and Reality	2013	Exploring Researcher Positionality and Cultural Interventions
P	Correa and Holbert [36]	Creating purpose: A case study of the creative process of an African American teen in maker education	2020	Exploring Researcher Positionality and Cultural Interventions

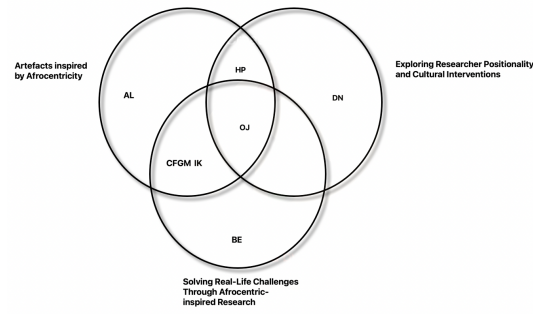


Figure 2: Thematic representation of reviewed literature. Source [owner]

4.3.2 Exploring Researcher Positionality and Cultural Interventions: Research papers **D**, **H**, **J**, **N**, **O**, and **P** offer insights into researchers' positionalities on Afro-centered research. Of all contributions, only the authors of **P** [36] and **J** [124] explicitly declare their epistemic stance for research, while the authors of **H** hint on Afrocentricity inspiring their work. These papers contribute to the corpus of knowledge on how the complexities of human experiences can be navigated, so as to design and/or introduce interventions that reflect their cultural experiences. In healthcare AI, **D** [9] explores the impact of Afrocentric ethics influenced by Ubuntu philosophy. This approach aims to address limitations faced by vulnerable groups through fairness, respect, and community engagement, resulting in a unique 5-pillar framework for AI system development. It advocates for diverse data collection and prioritizing safety over profit, fostering responsible and human-centric AI solutions.

"While many aspects of our personal and professional lives are structured by assumptions about individualism, co-authors 1 and 2 have extensive engagement and insights about communal relations in Africa, which grounds our definition of Ubuntu. An Ubuntu conceptualisation of personhood involves valuing communal relationships, and appreciating that identifying with others often involves a shared way of life" - **D** [9]

In a manner reminiscent of **D**, authors of **H** [45] have passionately crafted a comprehensive conceptual framework. This framework explores the complex relationship between indigenous African heritage, contemporary black influences termed Counter-Hegemonic Practices (CHP), and Computer Science education. Their approach aims to address issues of power and social justice overlooked by existing paradigms in computing education. The authors commendably establish a link between Computer Science education, the computational potential of CHP, and community development. Notably, they inclusively consider African descendants, even those not residing in Africa, who draw cultural influence from diverse African traditions, particularly African American communities. Their study also suggests promising applications of CHP in Computer Science education through their exploration of ethnocomputing. The authors in **J** [124] argue that outsiders' documentation and digitalization efforts have led to a misleading

portrayal of African cultural identity. This has resulted in the dissemination of inauthentic imagery and an inaccurate perception of indigenous African communities. They explore how technology, guided by Afrocentricity and Ubuntu principles, can support cultural preservation and empowerment among two Namibian communities (ovaHimba and San). The authors uncover issues like exotification of the ovaHimba tribe and romanticization of the San tribe, leading to the generation of three key themes (community reflections, digital self-representation, and digital empowerment). The study concludes with a thought-provoking video compilation that weaves together diverse content threads and reactions.

"Our approach to minimize problems of inappropriate technology and misrepresentations has throughout been to involve indigenous communities as equal partners in the design of the technologies supporting the safeguarding of their own cultural heritage" - **J** [124]

Papers **N** [51], **O** [76], and **P** [36] offer distinct perspectives on utilizing Afrocentricity for instigating change. **N** [51] explores the interdisciplinary aspects of human-machine interaction in social robotics through experimental psychology. It underscores psychology's role in HCI, emphasizing causal relationship challenges in technical systems and human contexts, promoting collaboration across disciplines, and considering dispositional and situational factors in social judgments and interactions. Contrastingly, **O** [76] primarily addresses misconceptions related to Information and Communication Technology for Development (ICT4D) implementation. The study examines community entry in ICT4D, the role of cultural interpreters during community engagements, and the appropriate responses of outside practitioners to rural communities' technology needs. The research paper emphasizes its contribution to knowledge regarding the introduction of ICT4D artefacts. The study concludes by critically assessing the interpretation of UNESCO's ICT-CST policy framework in the distinctive community context. In contribution **P** [36], the authors use multiple theoretical perspectives to study how an African American teenager created an Afrofuturist artefact through a critical constructionist design approach. They emphasize Asante's framework and an Afrocentric perspective to understand creativity, especially community impact [16]. The study highlights the importance of ethnicity for the second and third authors, leading to a focus on Black participants' experiences. Findings are examined through cognitive and socio-cultural

lenses, with the Afrocentric viewpoint considering emotions. The study effectively integrates Western and non-Western creativity paradigms, suggesting its value for appreciating diverse African creativity.

“A combination of participant observation and critical ethnography was used to study ongoing community entry practice and how ICT and ICT policy were introduced in the Happy Valley community.....” - **O** [76]

4.3.3 Solving Real-Life Challenges Through Afrocentric-inspired Research. Five review papers, **B,C,E,F** and **M** provide insight into Afro-centred approaches for solving real life challenges. While **B** and **E** do not explicitly disclose the positionalities and epistemic stance of the authors, **M** hints on the study being inspired by the principles of Afrocentricity and the authors of **C** and **F** clearly define Afrocentricity as the motivating paradigm for their work.

B [60] investigated the impact of risk assessment tools on U.S. criminal justice decision-making. Through their experimental study, they discovered that participants did not consistently use risk assessment information. Furthermore, disparities in predictions for black and white defendants were observed, indicating inequality. The study did not explore how judges’ assessments might be influenced by racial priming. Nevertheless, the authors made recommendations for better risk assessments and highlighted potential racial discrimination in the courtroom, aligning with their experimental findings. On the other hand, the authors of **C** [17] explored the impact of COVID-19 on Small and Medium-sized Businesses (SMBs) in Kenya, using Afrocentric and Ubuntu principles. Through interviews with 40 Kenyan SMBs, they examined how SMBs adapted to the pandemic using technology. The study argues against a one-size-fits-all tech approach, emphasizing the importance of African social-cultural norms like Ubuntu and Indaba. It introduces the concept of “socio-tecture” to explain Kenyan SMB practices, providing insights for HCI research on African SMBs. Authors of **E** [87] examined how Swaziland’s National ICT policy views development through qualitative analysis, using Habermas’s communicative action theory and interviewing 16 policy actors. They found the policy emphasizes Western standards, modernization, and economic competitiveness for development, neglecting the importance of an enabling technology environment. However, the study’s single-case focus limits generalizability, with potential insights for policymakers on technocentric development.

“..but instead from an afrocentric discourse where knowledge and innovation from the continent is legitimate and equal to others, and can spark innovation and problem-solving in similar contexts on and off the continent.” - **C** [17]

In study **M** [71], the authors aimed to create a practical tool for engineers to consider intercultural differences and inclusivity during problem definition in humanitarian design. They identified a literature gap in this area and proposed a comprehensive approach. The methodology involved analyzing existing toolkits and resulted in a problem analysis toolkit. In study **F**, authors applied Afrocentricity and Ubuntu principles to design tools for wildlife conservation. They used a mixed methods approach, selecting a web-based portal for audio-visual, text, and geolocation data. Collaborating

with indigenous communities using an Afrocentric paradigm, they designed tools as boundary objects. The co-design revealed diverse accuracy perspectives with implications for future conservation tools.

“We have deployed a community-based co-design (CBCD) approach... framed within an an Afrocentric research paradigm and Ubuntu-inspired epistemology” - **F** [93]

5 DISCUSSION

Understanding researchers’ positions is crucial to assess how their beliefs impact study design and outcomes [65, 117]. Beliefs, influenced by lived experiences, teachings, and contextual factors, are vital in fields like HCI, especially regarding Afro-centred research and potential influences tied to ethnical intersectionalities [61, 85, 119]. We explore insights, such as researchers’ unclear positions in HCI papers, a shared inclination for Afro-centred research, and the limited impact of ethnical intersectionalities on study design. We also emphasize the need to decolonize HCI and offer a modified Afro-centred paradigm.

5.1 The importance of clarifying a positionality

Positionality which differs from the term, “position”, is a complex concept that describes how the position of an individual in a particular context affects their perspectives and understanding of the world [23]. It highlights how the background, geographical location, privileges, identities, and socio-cultural influences of a person, shapes their understanding, decision making processes and subsequent interpretation of data. These are important tenets of reflexivity. Reflexivity refers to a deliberate self-scrutinizing process in which the researcher is conscious of how their biases and perceptions could possibly influence research and design practices [35]. The reflexivity of a researcher is significantly influenced by their positionalities. To provide more context to this discourse, imagine a Western researcher conducting Afro-centred research. We can be certain that there is a risk that the quality of study outcomes or even the design of the study could be questioned. That is because, regardless of how empathetic and enthusiastic this researcher is towards Afro-centred research, this person may have preconceived perceptions of the target audience (some of which may be colonial, exclusive and racist) [24, 111], have a different reflexive stance [3, 48], and differences in culture may result in a complete lack of understanding of the target population. Do we then exclude these researchers from this discourse? Absolutely not, thus the importance of proposing a framework as seen in later parts of this paper.

Within the context of this study, it was noted that researchers asserted that their studies were grounded in the promotion of authentic African experiences using different African concepts. Nonetheless, we observed that a consistent framework for this assimilation was absent. Furthermore, the precise standpoints of the scholars in relation to Afrocentrism or Afrocentricity remained unclear, contingent upon the interchangeable use of these terms. Instead, our observation reveals a pronounced focus on Afro-centredness and challenge of colonial imperatives, rather than the inclusion of the clear positionalities of the engaging scholars. Out of the 16 reviewed

papers, only 5 (C, F, I, J, and P) clearly declare their epistemic stances and positionalities. The adjective, “Afrocentric” is used frequently and ethical and philosophical principles such as *Ubuntu* are adopted or suggested. It is worth noting that while *Ubuntu* encourages communal inter-connectedness and interdependence, it predominantly finds its roots in the southern African context [75]. While we are not focused on the subject of pan-Africanism, the application of the concept of *Ubuntu* within various African contexts beyond the boundaries of Southern Africa presents as a noteworthy challenge depending on perspective and interpretation. Given the emphasis placed upon cultural specificity and contextualization, principles of this nature might not be uniformly generalizable. Consequently, this prompts us to ask questions regarding its use in domains such as AI for healthcare across cross-cultural borders as implied by D.

With regards to promoting Afro-centredness, if we assume that Afrocentricity is intended whenever the term Afrocentric is used [as implied by Asante], then Afrocentricity (regardless of its limitations) as a paradigm influences research and design practices. **The bone of contention** lies in the perception and understanding of this paradigm itself, as it raises several critical questions if it is used haphazardly [4, 78]. In his seminal manifesto, Asante [15] underscores the significance of acknowledging the rich African heritage. He champions the necessity for assuming authority over prevailing narratives and acknowledges the global dispersion of Africans - a consequence of the transatlantic slave trade. However, notwithstanding the commendable attributes of these postulations, it is imperative to critique their application in HCI. It is unclear how most of the researchers in the reviewed contributions situated themselves with respect to the current narrative of Afro-centred research. What was observed was their desire to change and control the African narrative [H, J, O], create change in African societies [E, I, K, P, C, E, M], and reduce racially induced bias [A, B, N,]. Because we cannot gauge the extent to which the reviewed contributions addressed the influence of ethnical intersectionalities in their work, it may be difficult to foster more inclusive and culturally sensitive HCI approaches that resonate with the multifaceted realities of users from diverse backgrounds [122, 133]. This challenge could have been simplified if Afro-centred researchers explicitly described their positionalities : if they just wanted to conduct research for indigenous Africans in Africa without a decolonial praxis, if they were inspired by Asante’s *Afrocentricity* or Afrocentrism, or if they were inspired by other decolonial paradigms that could promote Afro-centredness.

5.2 Conducting research for Africans - to decolonise HCI

Contemporary proposals in decolonisation tend to agree that colonial epistemologies are still being applied in African countries while neglecting indigenous knowledge systems and practices [20, 49, 50]. It is postulated that colonial education was not birthed out of the desire to create indigenous African elites, but to propagate colonial theologies and smoothen communication with the locales for business and governmental purposes [83, 98, 123]. Unfortunately, numerous African scholars have inadvertently absorbed these conceptions and epistemologies, thereby perpetuating colonial systems

without critically examining their authenticity or considering the potential for diverse perspectives [2, 11, 42, 67]. Thus, if indeed we must decolonise any aspect of education, even before design practices we must begin at grass root level i.e. African societies communities. Like Schultz et al. [118] and the authors whose papers we reviewed implied, the perspectives of indigenous Africans and colonized/marginalized societies must be considered, and an extra effort should be made to understand how their needs may differ from what colonialist perspectives think they should be. However, researchers, both African and non-African must go beyond observing lived experiences to understand cultural realities and how historical events have impacted the state of being [30, 132]. We argue that this has to be a conscious iterative process, not for the purpose of dispelling every colonial influence that may have practical benefits for African communities, but for posterity sake - knowing that whatever research and design approaches are being eliminated, assimilated and/or disseminated in African societies are intentionally promoting what Africans define as authentic to themselves. However, we infer that although Afrocentricity is a valid standpoint for research and design practices, the absence of a robust framework has led to variations in how studies are conducted, even among researchers who advocate for Afro-centred values. This lack of consistency hampers the reproducibility of findings. This is why we have called for an examination of the limitations and practical challenges of Afrocentricity in HCI. Such scrutiny can foster a more informed dialogue on how cultural frameworks influence technology design.

5.2.1 Revisiting Afrocentricity: A Critical Examination. Asante has been instrumental in advocating for Afro-centred paradigms and research methods, thereby serving as a catalyst for the inspiration and drive observed among Afro-centred HCI [16]. As the father of Afrocentricity, he antagonised colonial/neo-colonial research pathways and emphasized the need for African cultures to be revived and studied. Basically, researchers had to be submerged in the African experience by becoming familiar with every aspect of the African culture [89]. While Afrocentricity has inspired Afro-centred research and we do agree with some of its principles, it is not without limitations:

- (1) Afrocentricity is said to be inclusive of Africans in diaspora. However, there is an exclusionary risk among diasporic Africans who can neither relate to indigenous cultural practices nor visibly incorporate African concepts in their everyday living through language, actions, and values. The inability of these Africans to relate to their culture beyond secondary learning (i.e. cultural and identity empowerment through education) because of the absence of a lived experience does not infer mental subjugation to colonialism. However, it may lead to the potential invalidation of study outcomes from diasporic Africans by individuals who believe they have entirely lost their identities. This assumption arises from the belief that the study design and feedback processes are inherently “non-African” according to their own comparative standards with Africans in Africa.
- (2) Afrocentricity implies that Africans express negative traits such as being decentered due to colonialism [16], and thus does not give room for the appreciation of non-African/Western

influences that may have potential benefits to the African community.

- (3) There are opinions that may contradict what is defined as culturally authentic due to religious and other socio-cultural affiliations. Afrocentricity does not consider these opinions. These affiliations which may be colonial themselves may have become a strong part of the African identity, and such intersectionalities are not necessarily explored.
- (4) Ideally, culture should evolve to cater for the needs of the society [53, 68]. Although several colonial researchers have wrongly described African cultures as static and timeless [66, 99], we should consider the possibility that some unaltered aspects of African cultures might not be relevant today. Afrocentricity does not address this possibility and gives little room for cultural assimilation. Because it is perceived that cultural transformation, a perceived outcome of colonisation and cultural assimilation, could foster cultural genocide [101].
- (5) There is a recurring premise on centering on “blackness” that is promoted by Afrocentricity [38]. This is evident in Afro-centered research where there is often inadequate representation of Africans with features associated with Eurocentric or Middle-Eastern backgrounds. This limitation compounds the exclusionary nature sometimes associated with Afrocentricity.

5.2.2 Suggestions for a Decolonization paradigm to promote Afro-centred research and design practices. We propose a new viewpoint on Afrocentricity - **Afrocentric Postmodernism** (shortened as Afro-postmodernism), which promotes a culturally immersive intellectual African perspective while mitigating existing limitations in recent Afro-centred approaches to research. It can be described as a decolonised approach that rejects racism and other exclusionary factors while embracing cultural assimilation, dynamic growth, and if necessary some non-African concepts that have practical efficacy in African communities. Some of these non-African concepts may be of colonial origin, but our argument lies on their efficacy rather than the motivation behind their existence.¹ We also advocate for scholars to be able to differentiate between colonial concepts and non-African concepts that were used during the colonial era but were not of colonial origin. For example, in section 5.2.3, we introduce John Locke’s concept of *tabula rasa* (blank slate) in our Afro-postmodernist framework. While this philosophical concept has been misconstrued to justify certain colonial actions such as the belief that Africans were primitive and closer to the “blank slate” and thus should be “civilised” [55], it is not inherently colonial and predates classical colonisation [44]. The unfortunate association of John Locke with slave trade, should not drown the potential benefits of the *tabula rasa* theory [10, 25, 125].

¹For instance, there is a complex relationship between the assimilation of Christianity within several African communities and current perceptions of health needs [7, 106]. While certain Christian practices are indeed colonial in nature, some Christians claim that their belief system has contributed to improved mental health outcomes compared to traditionalists and atheists, even though some of the latter claim that some Christians visit them for powers [8, 18, 108]. If this complex and nuanced case represents the established research context, researchers would need to find a balance between their decolonial approach, understanding aspects of assimilated cultures or religions that have been negatively affected by colonialism, and respecting the positive outcomes of those belief systems

The term “Afrocentric Postmodernism” emerges from the fusion of the terms Afrocentric and Postmodernism. “Afrocentric” denotes a perspective that accentuates African culture, historical narratives, and lived experiences. In this discourse, it is not meant to reflect Afrocentricity. However, we use the term to communicate its relevance in representing African communities (at home and in diaspora). “Postmodernism”, though somewhat difficult to define, embodies the progressive capacity of humanity to be innovative and creative [1]. Jean-François Lyotard [82], contributes to current understanding of Postmodernism in his work *The Postmodern Condition: A Report on Knowledge* by encouraging diversity, promoting the understanding of nuances and the plurality of knowledge. It is worth noting that although the word “Postmodernism” presently lacks significant prevalence in characterizing HCI research and design practices, its legacy guides different aspects of life in this present day [54]. With Afrocentricity, there is also little room for cultural assimilation, appreciation and possible appropriation of non-African cultures. Thus, while not free from its limitations, its integration with the Postmodernist paradigm unveils nuances in socio-cultural experiences. This creates multiple avenues for design implications in HCI, and the plurality of thought - knowing that it is okay to change one’s mind due to the entrance of new information [21]. This is our attempt at finding a balance between Postmodernist approaches and Afrocentricity (see **Figure 4**). An important aspect of Afro-postmodernism resides in its deliberate negation of subjugatory, disempowering, and illusory conceptions previously imposed upon indigenous African contexts. Furthermore, the ideology of Afro-postmodernism includes Africans who may have found themselves marginalised due to their perceived divergence from supposedly established parameters of prescribed African traditions.

5.2.3 Step-by-step guide for Afro-postmodernist inspired research. While certain aspects of this step-by-step guide influenced by the Afro-postmodernist paradigm may be practiced by Afro-centred researchers, the difference lies in the intentional generation of insight from new found knowledge in that context which may contradict existing opinions of what authentic African experiences ought to be, encouragement of cultural assimilation, not treating African communities as the same, and giving room for the acceptance and integration of finite non-African concepts which may be of colonial origin but offer practical efficacy to the targeted African context. We also encourage the use of abductive analytic approaches using Afro-postmodernism as the conceptual framework for deductive conclusions.

- (1) Borrowing from John Locke’s *essay concerning human understanding* [109], we advocate for researchers to approach their studies without any preconceived notions or innate ideas about Africans. Although attaining a *blank slate* or *tabula rasa* seems unrealistic due to the accumulation of experiences during the course of a person’s life, being reflexively intentional about dumping implicit biases and not making any assumptions is a great way to start. Discussing reflexivity in this context is tricky, as the researcher must attempt to not dwell on existing preconceptions but learn like a new baby, and develop perceptions from new found knowledge.

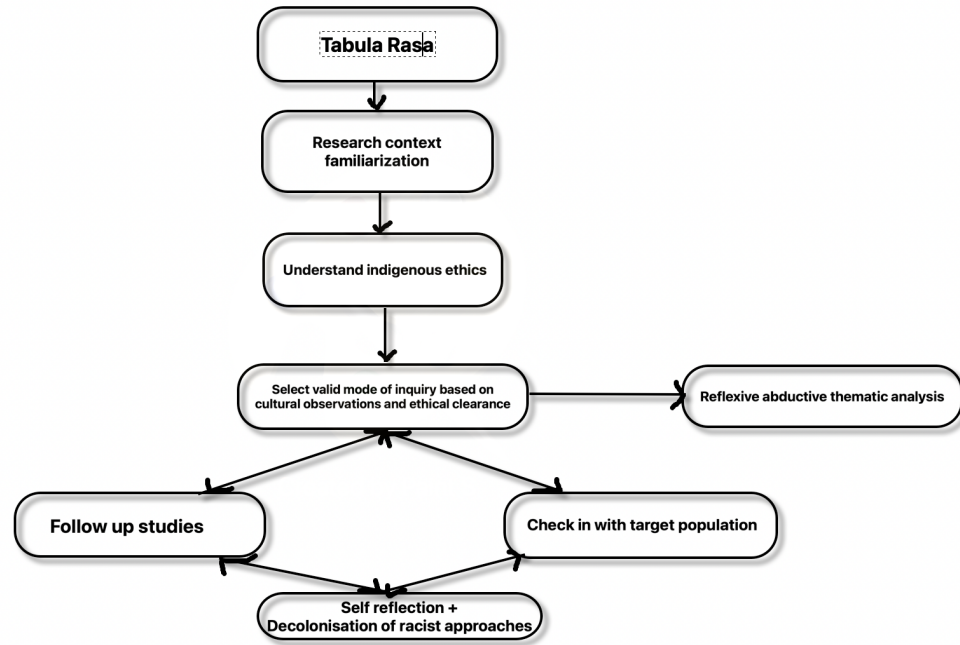


Figure 3: Suggested framework for Afro-centred research in HCI

- (2) As previously discussed, researchers must be intentional about being familiar with socio-economic, cultural and demographic influences on their research context. They must recognise the presence of intersectionalities and categorically define their objectives based on the integration of their preconceived ideas [i.e. the reason why they are conducting research in the first place (*do not confuse with the intentionality behind maintaining a blank slate*)] with context specific observations. They must dedicate this time to identify existing colonial constructs and sieve through to find relevant concepts (if any).
- (3) We propose the deconstruction of research methods to align with what Africans (in that context) would perceive as a valid mode of inquiry. This suggests that research approaches should be tailored to respect the cultural values of various African societies and should adhere to indigenous ethical principles. While obtaining official ethics approval is a necessity, it is important to recognise that there may be indigenous ethical norms and practices that may influence any stage of the study. Although obtaining approval of this form of ethics may be impractical depending on the context, it is important that researchers understand what to do and what not to do before progressing with their studies. E.g. Rejecting kolanut in a Nigerian Igbo setting is a sign of bad etiquette and may result in lack of access to the study population [31, 129].
- (4) After data collection, it is advised that researchers undergo a satisfactory cycle of validating their data and research method by conducting follow up studies, checking in with their target population or allied institutions and reflecting on self, ensuring that while they do not promote racism or any other -ism, they respect practical "non-African" approaches

and detours from cultural representations, as long as it is necessary or if the target population leans in that direction. Constructive arguments with members of a community should also be encouraged during design practices [41]. It will be interesting to observe that the voices of present day Africans reflect a myriad of concepts some of which may not be representative of their culture.

- (5) It is expected that the corpus of data obtained will be significantly rich, diverse, culturally representative, inclusive and modern. These can then be analysed using reflexive abductive thematic analytical approaches. Abductive reasoning involves a back and forth between what has been generated from the data and what the researcher thinks based on experiences from their learning period and a guiding conceptual framework. It is imperative that they establish a dynamic balance between inductive cognitive processes and deductive Afro-postmodernist logical frameworks. To facilitate this, themes should be generated inductively after a significant number of coding iterations. Then, they should either be aligned with existing literature or justified with suitable evidence culled from the study.

6 LIMITATIONS

- (1) Although we assume that this decolonial research praxis would be applicable in ethnographies, interviews, and case studies, other research methods may not benefit from our proposed framework. Thus, we advocate for an encompassing decolonial framework for other research approaches that may not fit into our narrative.

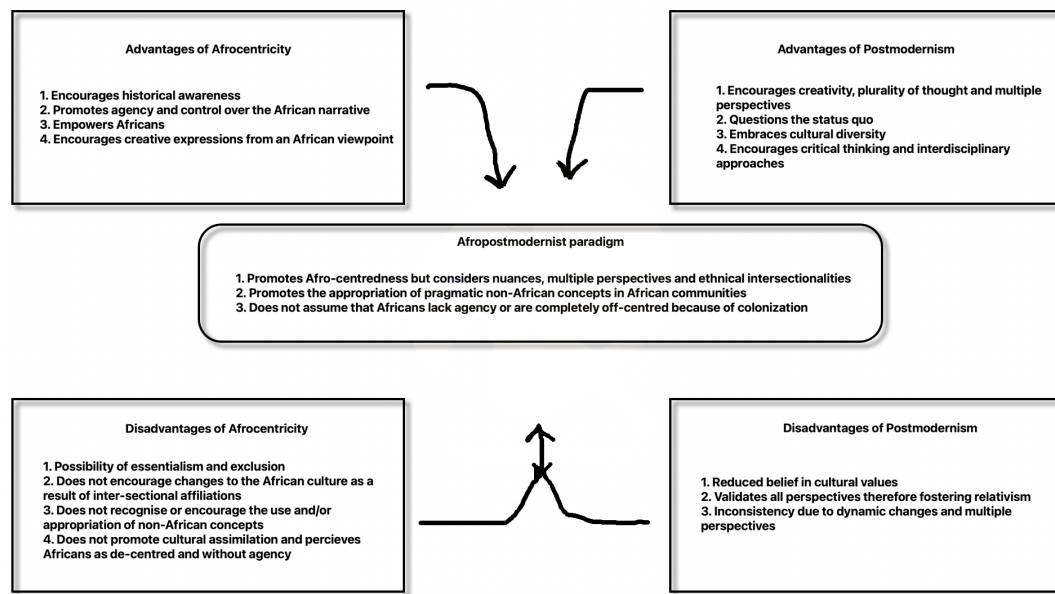


Figure 4: Integration of Afrocentric and Postmodernist paradigms. Source [owner]

- (2) By limiting the scoping review to peer-reviewed empirical research conducted within the last decade from specific databases, we may have left out information that may have influenced our research contribution. Thus, further reviews including broader databases and possibly grey literature could be conducted and compared to our findings.
- (3) Although we understand the potential friction between Afrocentricity and Postmodernism, we do not provide exhaustive information on Postmodernism in HCI as our focus was on Afrocentricity. For future work, we propose an exhaustive commentary of both paradigms and their effective use in HCI.

7 CONCLUSION

In this scoping review, we found a total of 16 empirical HCI contributions inspired by Asante's Afrocentricity. We aimed to uncover how it impacts research and design practices based on the positionalities of researchers who employ this lens. However, we also discovered a lack of uniformity in approaching Afrocentric research, which may affect their reproducibility. Going forward, we introduce Afropostmodernism and challenge the mindset of researchers inspired by Afrocentricity, encouraging them not to disregard Western theories entirely. We also advocate for a decolonization framework blending elements of both Afrocentric and inclusive Western approaches. Our proposition advocates for a balanced perspective that transcends colonial paradigms, fostering inclusivity and openness.

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